Name: .....

Category: Physics

1. The phenomenon responsible for the splitting of white light into its component spectrum when passing through a glass prism is:

   Options:

   A. Polarization
   B. Dispersion ✓
   C. Diffraction
   D. Total internal reflection

2. Three resistors of 2 ©, 3 ©, and 5 © are connected in series across a 20 V DC supply. Ca the circuit.

   Options:

   A. 1.5 A
   B. 2.0 A ✓
   C. 10.0 A
   D. 4.0 A

3. A body of mass 5 kg is accelerated at 4 m/s². What is the magnitude of the net resultant force acting on it?

   Options:

   A. 9 N
   B. 25 N
   C. 15 N
   D. 20 N ✓

4. Which of the following electromagnetic waves has the shortest wavelength and highest frequency?

   Options:

   A. Ultraviolet rays
   B. Microwaves
   C. X-rays
   D. Gamma rays ✓

5. What is the SI unit of electric capacitance?

   Options:

   A. Farad (F) ✓
   B. Henry (H)
   C. Tesla (T)
   D. Weber (Wb)

Category: Chemistry

6. What is the oxidation state of Chromium (Cr) in K2Cr2O7?

   Options:

   A. +7
   B. +5
   C. +6 ✓
   D. +3

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7. According to Boyle's Law, at constant temperature, the volume of a fixed mass of gas is:

Options:

A. Directly proportional to its pressure
B. Inversely proportional to its pressure ✓
C. Independent of pressure
D. Directly proportional to square of pressure

8. What is the pH of a 0.001 M solution of hydrochloric acid (HCl) assuming complete dissociation?

Options:

A. 4.0
B. 3.0 ✓
C. 1.0
D. 2.0

9. Which type of chemical bond is formed by the complete transfer of valence electrons from a metal to a non-metal?

Options:

A. Metallic bond
B. Covalent bond
C. Hydrogen bond
D. Electrovalent (Ionic) bond ✓

10. An element with atomic number 17 belongs to which group in the Periodic Table?

Options:

A. Noble gases (Group 18)
B. Halogens (Group 17 / VIIA) ✓
C. Alkaline earth metals (Group 2)
D. Alkali metals (Group 1)

Category: Biology

11. The largest part of the human brain responsible for voluntary actions, memory, and intelligence is the:

Options:

A. Cerebrum ✓
B. Cerebellum
C. Hypothalamus
D. Medulla oblongata

12. Which hormone secreted by the islets of Langerhans lowers blood glucose concentration?

Options:

A. Insulin ✓
B. Glucagon
C. Thyroxine
D. Adrenaline

13. Which blood group is universally recognized as the universal red blood cell donor?

Options:

A. Group AB Rh+
B. Group A Rh+
C. Group B Rh-
D. Group O Rh- ✓

❤ from Quemint

14. Deficiency of Vitamin K in the human diet primarily leads to:

Options:

A. Delayed clotting of blood ✓
B. Scurvy
C. Rickets in children
D. Night blindness

15. Which plant tissue conducts manufactured organic food from the leaves to storage organs?

Options:

A. Phloem sieve tubes ✓
B. Cambium ring
C. Sclerenchyma fibers
D. Xylem vessels

16. In human circulation, oxygenated blood returns from the lungs to the heart through the:

Options:

A. Superior vena cava
B. Pulmonary vein ✓
C. Pulmonary artery
D. Coronary sinus

17. During mitosis, chromosomes align along the equatorial plate during which phase?

Options:

A. Metaphase ✓
B. Telophase
C. Prophase
D. Anaphase

18. Gaseous exchange in the lungs of mammals takes place across the walls of the:

Options:

A. Bronchioles
B. Trachea
C. Alveoli ✓
D. Pleural membrane

19. The functional unit of the mammalian kidney responsible for ultrafiltration and selective reabsorption is the:

Options:

A. Axon
B. Alveolus
C. Nephron ✓
D. Sarcomere

20. Which cellular organelle is primarily responsible for oxidative phosphorylation and ATP synthesis?

Options:

A. Mitochondrion ✓
B. Lysosome
C. Golgi apparatus
D. Endoplasmic reticulum

Category: English

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21. Had I known you were arriving today, I _____ met you at the terminal.

Options:

A. would
B. will have
C. had
D. would have ✓

22. The suspect was exonerated after new forensic evidence emerged. "Exonerated" means:

Options:

A. Transferred to prison
B. Convicted swiftly
C. Interrogated harshly
D. Cleared of blame ✓

23. Choose the option that has the same vowel sound as the underlined letters in "PEACE":

Options:

A. Dead
B. Sieve
C. Great
D. Key ✓

24. Select the correctly spelt word:

Options:

A. Accomodation
B. Acommodation
C. Acomodation
D. Accommodation ✓

25. What does the idiom "to burn the midnight oil" signify?

Options:

A. To wake up before dawn
B. To instigate a protest
C. To study or work late into the night ✓
D. To waste household fuel

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